

Sendrow Build Pipeline

From Masao → Malachi. Last updated July 10, 2026 — added research findings from CARB, PACT, CDP, EcoVadis, and assurance-standard review.

This is the single source of truth for what to build and in what order. Work top to bottom. Don't skip ahead to a lower phase unless you're blocked on everything in the current one. If something in here conflicts with an old doc or Slack message, this doc wins.

How to read the status tags

- **VERIFY** — We think this exists or partially exists in the prototype. Your job: test it end-to-end as a real user would, note what's broken or missing, then fix. Don't rebuild from scratch if it works.
- **BUILD** — Doesn't exist yet. Build it.
- **IMPROVE** — Exists but needs polish per the notes.
- **DONE** — Confirmed working. Leave it alone unless a later feature touches it.
- **LATER** — Explicitly do not build now, no matter how fun it sounds.

Ground rules (read before writing any code)

1. **Config-driven formats, always.** Every output format (SB 253, CDP, buyer questionnaires) must be a template/config the system reads — never hardcoded logic. This one rule is what makes #9, #33, #35, and the future EPR module (#50) possible. If you find yourself writing an if (format == "CDP") branch, stop.
2. **Nothing shared externally except approved snapshots.** Live data never leaves a workspace. Every external share = snapshot + recipient + format + supplier permission, logged.
3. **Audit trail on everything from day one.** Every create/edit/approve/share event gets written to the immutable log (#13). It's much cheaper to log everything now than to retrofit later.
4. **Client separation is absolute.** Any query that could return another client's data is a company-ending bug. Write tests for this specifically.
5. **PACT V3 only — no exceptions.** PACT deprecated V2.x of its Technical Specifications as of April 1, 2026; V3 was published April 30, 2025 and is the only version worth building against. Use the PACT V3 OpenAPI schema and Simplified Data Model (Excel) as the reference — don't build our own shape and reconcile later. Long-term, register for the PACT Conformance Tool so we can be listed as a PACT Conformant Solution; that's free credibility with consultants and buyers.
6. **One feature = one branch = one checklist item.** When you finish something, mark it in this doc with the date. When you verify something, write one line on what you tested and what state it's in.
7. **Staging first.** Nothing ships to the live Render deployment without being tested on staging with the demo workspace data. No exceptions once we have a design partner in the system.

PHASE 1 — Verify and harden the core loop

Goal: the full journey — consultant sends request → supplier responds via magic link → evidence attached → consultant reviews/approves → snapshot frozen — works flawlessly with zero babysitting. This is what every demo runs on.

Order	Feature	Status	What it is + instructions
1.1	#1 Data Request	DONE — regression test	Consultant picks company, data types, time period, deadline. Confirm all four fields save and display correctly on the supplier side.
1.2	#2 Magic-Link Response	DONE — regression test	THE feature. Test: link opens on mobile, link works after 30 days, link is single-supplier scoped, expired/revoked links fail gracefully with a "request a new link" button.
1.3	#4 Guided Forms	IMPROVE	Upload page needs to be more digestible. Break long forms into one-question-per-screen or clearly grouped sections with a progress bar. Every question in plain English with an example ("Upload your last 12 electricity bills — a PDF from SCE's website is perfect"). Test with someone non-technical (ask Stella to run through it cold; if she asks a question, that's a bug

Order	Feature	Status	What it is + instructions
			in the copy).
1.4	#5 Evidence Attachment	DONE — regression test	Any number can have a bill/invoice stapled to it. Verify: multiple files per line item, files viewable in one click from the number, files carried into snapshots.
1.5	#7 Review & Approve	DONE — regression test	Verify approval writes who/what/when to the audit log and locks the record from silent edits.
1.6	#8 Snapshots	DONE — regression test	Approved data freezes into a locked, dated version. Verify snapshots are truly immutable (try to mutate one via the API, not just the UI).
1.7	#10 Sharing Permissions	DONE — regression test	Each share = this snapshot, to this company, in this format, with supplier's explicit OK. Verify recipient A can't discover recipient B's shares.
1.8	#13 Full Audit Trail	DONE — regression test	Confirm every event type from 1.1–1.7 actually appears in the log, the log can't be edited or deleted, and it's exportable as a readable report (auditors will want a PDF/CSV, not database rows).
1.9	#12 Prefill From Last	VERIFY	New requests pre-fill

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	Time		from the supplier's last approved data. Verify: prefilled values are visibly marked "from your 2025 submission — confirm or update" so nobody blindly resubmits stale numbers.
1.10	#6 Comment Threads on Data	VERIFY	Comments attach to the specific line item, notify the other party by email, and survive into the record's history. If it doesn't exist, build a minimal version: comment box per line item + email notification. Nothing fancier.
1.11	NEW — Save & Resume	BUILD	Auto-save the supplier's partial answers on the magic-link form every few seconds. Closing the tab and reopening the link returns them exactly where they left off, with a "3 of 7 sections complete" indicator. Suppliers fill these out in stolen moments; losing work once = they never come back.
1.12	NEW — Estimate vs. Actual Data Status	BUILD (small)	Every value can carry a status flag: "estimated" or "actual." When a supplier or consultant later replaces an

Order	Feature	Status	What it is + instructions
			estimate with the real number, log it as a correction (feeds #11 Restatement Alerts) rather than a silent edit. This mirrors how real enterprises operate today — e.g. METRO's own KPI reports disclose that late-arriving utility bills force them to estimate the final quarter and true it up the following year. Cheap to add now, painful to retrofit later.

PHASE 2 — Demo & design-partner readiness

Goal: I can put a real consultant (Kerri) into the product without embarrassment, and I can demo it to anyone on 5 minutes' notice.

Order	Feature	Status	What it is + instructions
2.1	#49 Demo Workspace	BUILD	One permanently seeded fake consultant account ("Pacific Sustainability Advisors") with 3 fake clients at different stages: one just requested, one mid-response with comments, one fully approved with a snapshot. Realistic numbers, realistic fake utility bills. Write a reset script so I can restore it to clean state before every demo. Do this early — every sales

Order	Feature	Status	What it is + instructions
			conversation depends on it.
2.2	#19 Multi-Client Dashboard	VERIFY	One login, all clients, status at a glance: who's behind, what's awaiting review, what's due. Verify statuses update in real time and the "what's due" sort is correct.
2.3	#20 Client Separation Walls	VERIFY + TEST	Write automated tests that attempt cross-client data access through every endpoint. This is the one feature where "seems fine" isn't good enough.
2.4	#22 White-Label Requests	BUILD	Consultant uploads logo + sets display name once in settings; all outgoing request emails and magic-link pages wear their branding. Keep "Powered by Sendrow" small in the footer. Scope: logo, name, accent color. Not custom domains yet.
2.5	#23 Engagement Templates	BUILD	"Save this request setup as a template" + "start new request from template." A consultant's standard SB 253 package becomes one click for every new client. Store as config

Order	Feature	Status	What it is + instructions
			(Ground Rule #1).
2.6	#21 Automatic Chasing	VERIFY	Scheduled reminder emails as deadlines approach (e.g., 7 days out, 2 days out, day-of, overdue). Verify: consultant can adjust cadence per request, reminders stop instantly on submission, tone is polite. Log every reminder sent so the consultant can see the chasing happened.
2.7	#46 Client Profile Basics	BUILD (small)	When any client/supplier is created, capture industry code (NAICS dropdown) and employee-count size band. Two fields, an afternoon of work, and it's the prerequisite for benchmarking later. Cheapest future-proofing we'll ever do.
2.8	NEW — Response Notifications	BUILD (small)	Email the consultant the moment a supplier submits or comments, with a deep link straight to the item. Consultants shouldn't have to poll the dashboard.

PHASE 3 — The moat: reshaping and the format engine

Goal: "answer once, output many formats" actually works. This is the core product insight from the Kerri call and our sharpest differentiation. It deserves the most careful engineering in the whole roadmap.

Order	Feature	Status	What it is + instructions
3.1	#9 Reshaping	VERIFY, likely rebuild-as-config	One approved snapshot → SB 253 format, CDP-style report, custom questionnaire, or plain Excel, one click each. If the current version has any hardcoded format logic, refactor now into the template engine before adding more formats. Ship with 3 formats: SB 253 structure, generic Excel export, and one real buyer questionnaire (I'll get an example from Kerri). Update: CARB published a real draft SB 253 reporting template in October 2025 — Excel-formatted, with free text and dropdown fields covering org info, third-party verification, inventory boundary, emissions disclosures, methods, and base year emissions. It's voluntary for 2026 but CARB plans to release a revised draft for public comment in summer 2026 and intends to make it mandatory

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			starting 2027. Build the SB 253 format against this real template, not a guess — I'll get you the current draft.
3.2	#35 Format Mapping Builder	BUILD	Internal admin tool: upload a new questionnaire, map each of its questions to a Sendrow data field once, save as a reusable template. This is how the format library grows without engineering work. V1 can be admin-only and ugly — it just has to make adding a format an afternoon of clicking instead of a code change. Important: CDP's 2026 questionnaire routes each company only to the questions that apply to it based on size, activity, and opted-in modules (e.g. opting into "water" pulls in a whole extra module). The mapping builder needs to support conditional/branching questionnaires, not just flat ones, or CDP mapping won't work.
3.3	#33 Maintained Output Templates	PROCESS + BUILD (small)	Templates need versioning: when CARB or CDP changes a form, we update the template

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			centrally and every future export uses the new version, while old snapshots record which template version they were exported with. Build the version field now; I'll own watching for regulatory changes. CARB's summer 2026 revised SB 253 template is our first real-world test of this — treat it as a fire drill for the versioning system.
3.4	#26 "Answer Once, Share Many"	BUILD	Supplier-side flow: when a second customer requests data the supplier already has approved, show "You already have this — share your existing snapshot in their format?" This is the fall-in-love moment and it's mostly UI on top of 3.1 + #10.
3.5	NEW — Duplicate Request Detection	BUILD (small)	The system-side half of 3.4: when a new incoming request overlaps data the supplier already has approved (same period, same categories), automatically flag it and offer the snapshot-share path instead of a blank form. Nobody should

Order	Feature	Status	What it is + instructions
			ever type the same number twice.

PHASE 4 — Supplier trust & stickiness

Order	Feature	Status	What it is + instructions
4.1	#24 Free Supplier Account	BUILD	After responding via magic link, one-click "claim your free account" (email + password, that's it) so suppliers can view their own data anytime. Never gate responding behind signup.
4.2	#37 Supplier Attestation	BUILD (small)	Before any external share: "I confirm this data is accurate to the best of my knowledge" checkbox, recorded with name, date, snapshot ID in the audit log. One button; auditors and buyers treat it as a real signal.
4.3	#38 Share Receipts	BUILD (small)	Log when a shared snapshot is viewed/downloaded by the recipient and show it to both supplier and consultant. Closes the loop on sharing.
4.4	#25 Data Ownership & Export	BUILD	"Download all my data" button for suppliers — everything, anytime,

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			in PACT V3 format plus plain CSV. The ability to leave is why they trust us.
4.5	#3 Reply-by-Email	VERIFY	Supplier replies to the request email with text or attachments and it lands on the right request automatically. If the current version is unreliable, a fine V1 is: attachments land in an "unfiled inbox" on the request for the consultant to slot in. Perfect parsing is not required; not losing anything is.
4.6	NEW — Section Delegation	BUILD	On the magic-link form, "send this section to a colleague" — supplier enters an email, that person gets a scoped magic link to just that section (e.g., facilities manager gets the utilities questions). The office manager rarely has every answer; today they forward the whole spreadsheet and chaos ensues. This kills that.
4.7	NEW — Deadline Extension Request	BUILD (small)	A "request more time" button on the supplier form that pings the consultant for one-click approve/deny. Beats

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			silent lateness and gives the consultant visibility.
4.8	#27 Claimable Starter Profile	BUILD	Invisible Efficiency Calculator output becomes a draft Sendrow record the supplier can claim with one click. Coordinate with me on the calculator side — don't start until the calculator's data model is settled.
4.9	NEW — Supplier Mini-Report Generator	BUILD	Once a supplier's snapshot is approved, one click produces a short (5–8 page) branded PDF sustainability report — likely the first one they've ever had. This extends #25/#27, gives suppliers something they actually want to share, and doubles as free marketing every time it circulates. Basic Module of the EU's VSME standard (11 disclosures, B1–B11) is a good skeleton to borrow structure from even though we're not targeting EU compliance directly.

PHASE 5 — Audit-grade depth

Order	Feature	Status	What it is + instructions
5.1	#17 Both Calculation Methods	VERIFY	Spend-based (rough) and activity-based (accurate) methods both supported, every number labeled which method produced it, and a summary stat: "X% of this footprint is activity-based." CARB has explicitly proposed both a spend-based and an activity-based method for Scope 3 in its 2027 rulemaking — this labeling isn't optional, it's regulatory language.
5.2	#14 Methodology Labels	VERIFY	Every calculated number tagged with the formula and emission-factor version used (e.g., "EPA 2025 factors, v2.1"). Store factor sets as versioned data, never constants in code. Assurance providers specifically test whether a factor is relevant (matches the actual activity), recent (current database version), and geographically correct — build the label schema to capture all three, not just a version number.
5.3	#15 Trust-Level Badges	VERIFY / BUILD	Every dataset labeled: self-reported / consultant-reviewed / assured. Badge

Order	Feature	Status	What it is + instructions
			travels with the snapshot into every share and export. Note: badges may need to apply at the individual-metric level, not just dataset-level — real reports (e.g. METRO's) mark specific KPIs as unaudited even when the rest of the report carries formal assurance.
5.4	#11 Restatement Alerts	VERIFY / BUILD	When corrected data is re-approved, everyone who received the old snapshot gets an automatic notification with a "what changed" diff. Depends on #8 + #10 share records being solid, and now also on 1.12 (estimate vs. actual status) — a value moving from "estimated" to "actual" should trigger this same flow.
5.5	#39 Factor Update Recalculation	BUILD	When new emission factors publish, one click previews a client's numbers recalculated — while old reports stay frozen on their original factor version. Falls out naturally if 5.2 versioning is done right.

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5.6	#18 Vendor Memory	VERIFY	Per-consultant vendor→category mappings remembered and auto-applied next time. Cross-platform smart suggestions are LATER — per-workspace memory only for now (privacy).
5.7	#36 Historical Data Import	BUILD	Importer for a consultant's existing client spreadsheets: upload Excel/CSV, map columns to Sendrow fields, import as prior-year data (marked "imported"). The single biggest adoption unlock. V1 can be semi-manual — you run it with the consultant on a call. Build the polished self-serve version only after we've done it manually 3 times and know the patterns.
5.8	NEW — Inventory Management Plan (IMP) Generator	BUILD	Auto-generate a formal IMP / methodology manual document from a client's stored methodology labels (5.2), organizational boundary settings, and factor sources. Assurance guides consistently list a documented IMP as

Order	Feature	Status	What it is + instructions
			the first thing an auditor asks for — this turns data we already store into a deliverable, basically for free once 5.2 is solid.

PHASE 6 — Consultant retention & upsell engine (after first design partner is live)

Order	Feature	Status	What it is + instructions
6.1	#44 Compliance Calendar	BUILD	Preloaded CA/CDP deadlines by client type + custom dates, feeding the dashboard's "what's due." I maintain the regulatory dates; you build the plumbing. Correction to keep in mind: SB 253's first Scope 1/2 deadline is August 10, 2026, not November — make sure this is right in the calendar and anywhere else it's referenced.
6.2	#45 Monthly Client Digest	BUILD	Auto-generated monthly summary per client (data received, flags raised, completeness %) the consultant can forward as-is. This justifies their retainer every month.
6.3	#43 Consultant Commentary Blocks	BUILD (small)	An "analysis" text section in every

Order	Feature	Status	What it is + instructions
			deliverable/export so the consultant's judgment is visibly part of the product.
6.4	#40 Hotspot Report	BUILD	On footprint approval, auto-generate top 5 emission sources with standard reduction suggestions — the consultant's ready-made pitch for their next engagement.
6.5	#41 Year-over-Year Narratives	BUILD	With two years of data, draft the explanation paragraphs from template sentences ("Scope 2 fell 8%, driven by...") for the consultant to edit. Templates, not AI.
6.6	#42 Score-Gap Flags	BUILD	Check client data against CDP/EcoVadis scoring rules, flag where points would be lost. I'll source the scoring rubrics. Two concrete hooks to build against: CDP's 2026 SME questionnaire now allows an SME to reach an "A" score for the first time (previously capped at B) — a clear before/after selling point. EcoVadis scores almost entirely on attached evidence

Order	Feature	Status	What it is + instructions
			per option (up to 55 supporting documents per assessment) — our evidence-attachment architecture (#5) is a natural fit for auto-assembling an "EcoVadis evidence pack" within that document budget.
6.7	NEW — Completeness Meter	BUILD (small)	Per-client progress bar: % of requested data received, % with evidence, % reviewed. Feeds the dashboard, the digest (6.2), and buyer analytics later. Build the underlying metric once, reuse everywhere.

PARALLEL TRACK — ongoing, not a phase

Item	Status	Notes
#31 PACT-Compatible Data Format	ARCHITECTURE RULE	Not a feature to "build later" — it's how we shape the data model from now on. Build against PACT V3 specifically (see Ground Rule #5) — V2 is deprecated as of April 2026.
#32 SOC 2 / Assurance Readiness	START PROCESS	We can't afford the audit yet, but adopt the practices now: access controls, encrypted secrets, dependency updates, incident log, backups tested. Also worth knowing: the GHG assurance standard most CARB guidance points to (ISAE 3410) is being

Item	Status	Notes
		withdrawn as of December 15, 2026, replaced by ISSA 5000 — use that term in anything forward-looking so we don't sound out of date. Keep a running "SOC 2 / assurance readiness" checklist so certification later is paperwork, not rework. I'll research timing/cost.
#48 Referral Routing	SPREADSHEET	I run this in a spreadsheet. Do not build software for it until volume forces us to.
NEW — VSME Format Watch	WATCH	The EU's Voluntary Sustainability Reporting Standard for SMEs (VSME) is now, as of the March 2026 Omnibus I Directive, a legal ceiling on what large EU companies can demand from SME suppliers. If any of our CA mid-market clients also sell into Europe, VSME becomes a real format-library candidate, not just an inspiration for the mini-report (4.9). Revisit if a design partner mentions EU exposure.

EXPLICITLY LATER — do not build yet

- **#16 AI Suggestions** — not until the human workflow is proven. When we do: AI suggests, human always confirms.
- **#28–30 Buyer features** (bulk campaigns, snapshot inbox, response analytics) — Phase 2 of the business, 2027. The snapshot/share architecture we're building now is what makes these easy later.
- **#34 Integrations** (QuickBooks, utility APIs) — email + upload covers v1.
- **#47 Peer Benchmarking** — needs volume; #46 in Phase 2 is the only prerequisite work. Never show comparison groups under ~8 companies.
- **#50 EPR Module** (SB 54 packaging) — second market, same engine. The only thing it needs from us today is Ground Rule #1.

Logistics so nothing gets mixed up

1. **Definition of done:** works on staging, tested on mobile, event logged to audit trail, demo workspace still functions, marked in this doc with date + one-line note.
2. **Verification format:** for every VERIFY item, add a line under it in this doc: "Tested [date]: [works / broken because X / doesn't exist]." Decide fix-vs-rebuild together with me before rebuilding anything.
3. **Weekly sync:** we review this doc together once a week — what moved, what's blocked, whether the order still makes sense. Order changes happen in the sync, not silently.
4. **When blocked on me** (copy, example questionnaires, regulatory details, Kerri input): tag it "BLOCKED — Masao" in the doc and move to the next item. Don't sit idle and don't guess on regulatory specifics.
5. **Scope discipline:** if a feature is taking more than ~1.5x its rough estimate, we cut scope in the weekly sync rather than gold-plating. Small ugly version shipped beats perfect version pending.